MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Helpline: +91-8171546558

**Subject:** Mathematics
**Class:** Class X (CBSE)

**CBSE CLASS 10 – PERIODIC TEST 08**
**Chapter 13: Statistics**

**Time:** 60 Minutes
**MM:** 30

**Student Name:** _____ **Roll No.:** _____ **Date:** _____

**GENERAL INSTRUCTIONS:**

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer (LA) questions of 5 marks each.
6. **Section E** comprises 1 Case Study based question of 4 marks with internal subparts.
7. Use of calculators is not permitted. All rough work and calculation steps must be clearly shown.

## SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

**Q1.** The mean of the squares of first five natural numbers is: [1]
(A) 11 (B) 10 (C) 12 (D) 15

**Q2.** In the formula Median $= l + \left(\frac{N/2 - cf}{f}\right) \times h$, $cf$ is the cumulative frequency of the: [1]
(A) median class
(B) class preceding the median class
(C) class succeeding the median class
(D) modal class

**Q3.** If the ratio of mean and median of a certain data is 2 : 3, then the ratio of its mode and mean is: [1]
(A) 3 : 2 (B) 5 : 2 (C) 2 : 5 (D) 7 : 2

**Q4.** For the following grouped data, the modal class is: [1]

| Class | 10–15 | 15–20 | 20–25 | 25–30 | 30–35 |
|---|---|---|---|---|---|
| Frequency | 30 | 35 | 20 | 40 | 15 |

(A) 15 – 20 (B) 20 – 25 (C) 25 – 30 (D) 10 – 15

**Q5.** If the mean of observations $x, x+3, x+5, x+7, x+10$ is 9, then the mean of last three observations is: [1]
(A) 10.33 (B) 11.33 (C) 12 (D) 10

**Q6.** **Assertion (A):** In finding the mean of grouped data by step-deviation method, the class size $h$ must be uniform across all classes.
**Reason (R):** The step-deviation variable $u_i = \frac{x_i - a}{h}$ requires a single common division factor $h$. [1]
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

## SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

**Q7.** If Mode – Median = 12, find the value of Median – Mean. [2]

**Q8.** If the mean of 10 observations is 25 and that of another 15 observations is 35, find the combined mean of all 25 observations. [2]

## SECTION C: Short Answer Type (2 × 3 = 6 Marks)

**Q9.** Find the mean number of defective components per batch from the given records: [3]

| Defects | 0–4 | 4–8 | 8–12 | 12–16 | 16–20 |
|---|---|---|---|---|---|
| Batches | 3 | 7 | 12 | 10 | 8 |

**Q10.** Calculate the mode of the following distribution of scores: [3]

| Class | 20–30 | 30–40 | 40–50 | 50–60 | 60–70 |
|---|---|---|---|---|---|
| Frequency | 25 | 40 | 42 | 33 | 10 |

## SECTION D: Long Answer Type (2 × 5 = 10 Marks)

**Q11.** The median of the following frequency distribution is 28.5. Find the values of $x$ and $y$, if the total frequency is 60: [5]

| Class Interval | 0–10 | 10–20 | 20–30 | 30–40 | 40–50 | 50–60 |
|---|---|---|---|---|---|---|
| Frequency | 5 | $x$ | 20 | 15 | $y$ | 5 |

**Q12.** Compute the Mean, Median, and Mode for the following frequency distribution: [5]

| Class Interval | 10–20 | 20–30 | 30–40 | 40–50 | 50–60 | 60–70 |
|---|---|---|---|---|---|---|
| Frequency | 4 | 8 | 10 | 12 | 6 | 4 |

## SECTION E: Case Study Based Question (1 × 4 = 4 Marks)

**Q13. CASE STUDY: MUNICIPAL PUBLIC HEALTH BLOOD PRESSURE SCREENING INITIATIVE** [4 Marks]

A metropolitan civic body conducted an adult cardiovascular health checkup camp, measuring the systolic blood pressure (mm Hg) of 80 visiting citizens.

| **Systolic BP (mm Hg)** | 100–110 | 110–120 | 120–130 | 130–140 | 140–150 | 150–160 |
|---|---|---|---|---|---|---|
| **Number of Citizens** | 8 | 14 | 25 | 18 | 10 | 5 |

(i) Find the median systolic blood pressure of the screened citizens. [2]

(ii) Identify the modal blood pressure class and state its lower boundary. [1]

(iii) How many citizens were identified with stage-1 or stage-2 hypertension (BP $\geq$ 140 mm Hg)? [1]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★